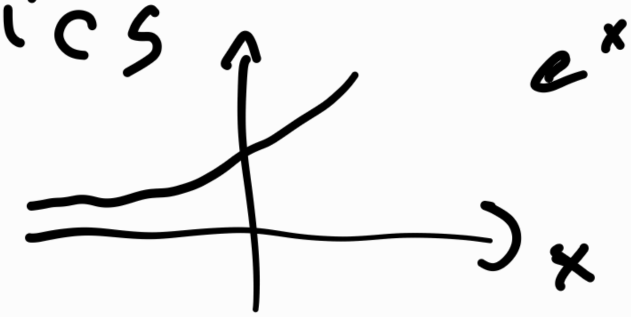
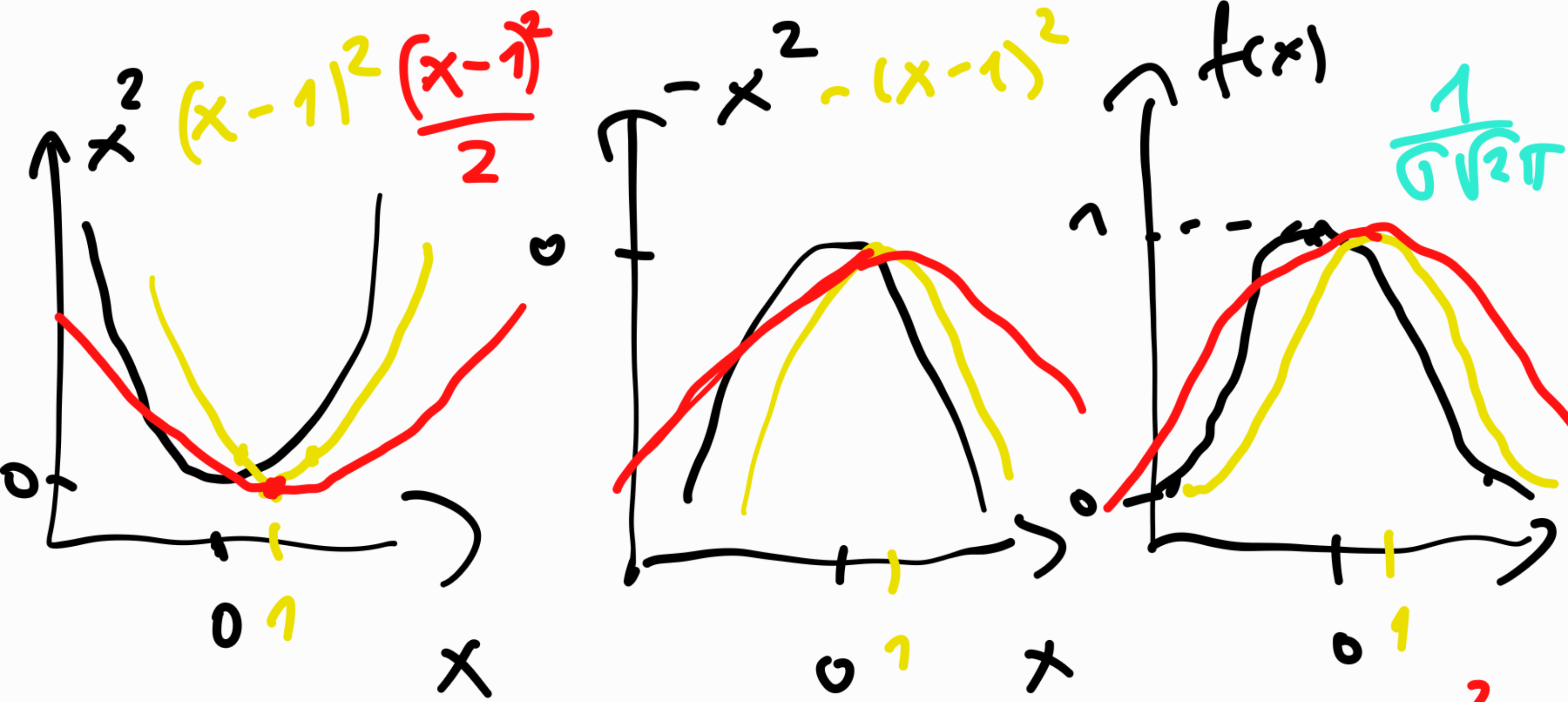
Intro to Statistics

$$f(x) \approx e^{-x^2}$$





$$f(x) = e^{-(x-\bar{x})^2}$$

$\bar{x} = 1$

$$f(x) = e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$$

$\sigma = 1$

$\bar{x}, \mu$ – mean [stredná hodnota]

σ – standard deviation (smerodajná odchýlka)

σ^2 – variance (rozptyl)

Data follows normal (Gaussian) distribution

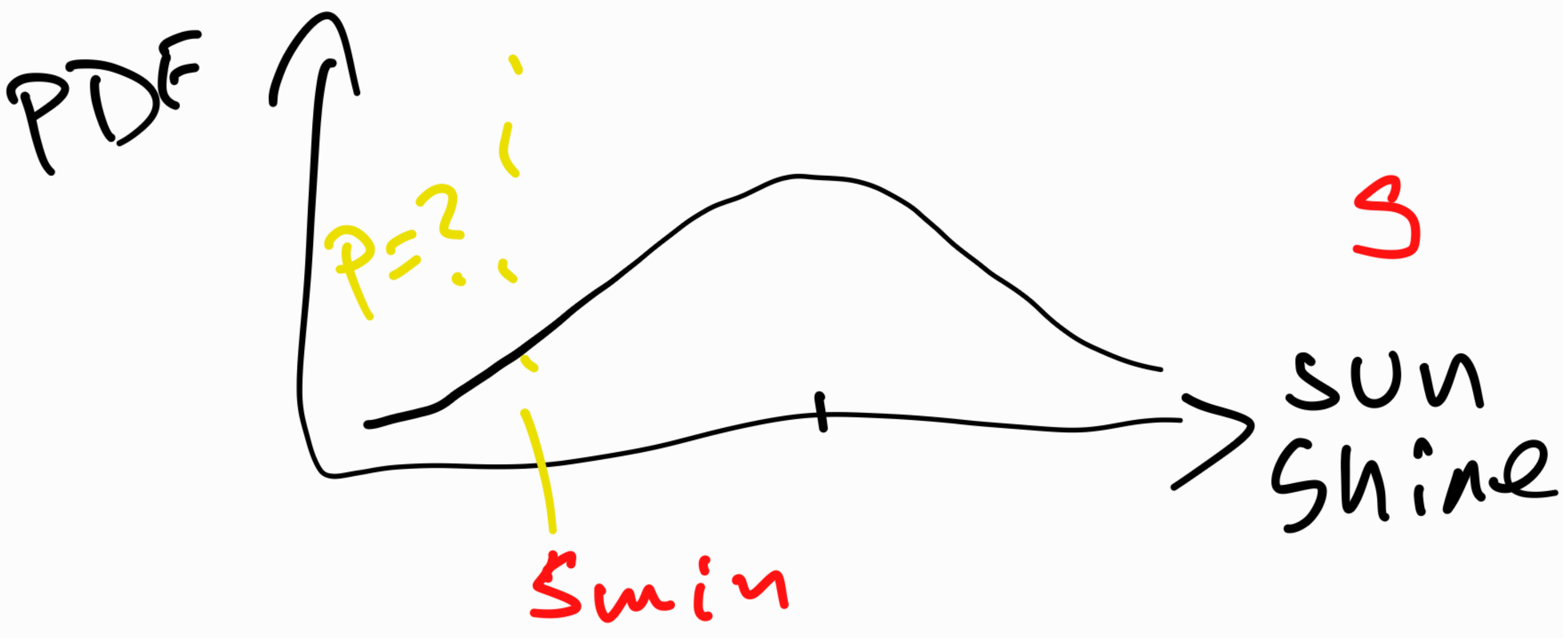
$x \sim N(\bar{x}, \sigma^2)$

with parameters $\bar{x}$ and σ^2.

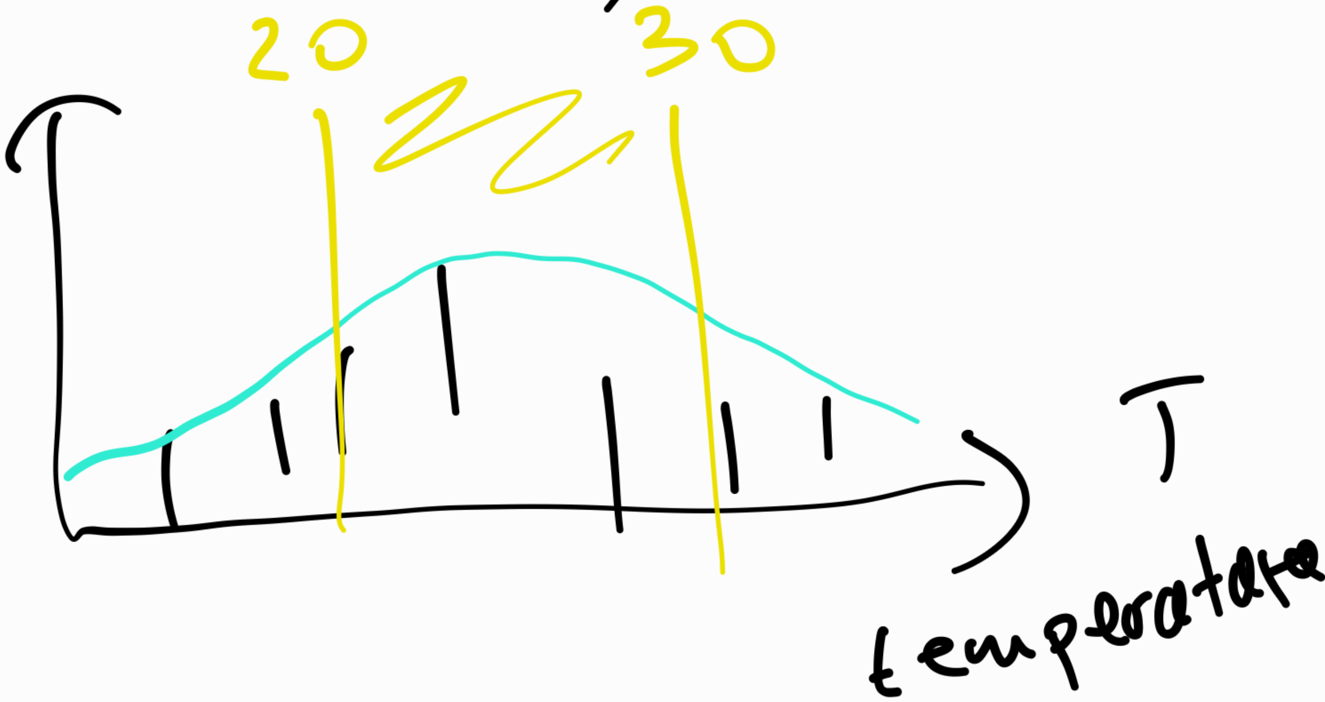
(pravdepodobnosti)

Probability Density Function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$$



$P(S \le S_{min}) = ?$



$P(20 \le T \le 30) = ?$

(distribučná)

Cumulative Distribution Function (CDF)

$$F(z) = P(x \le z) = \int_{-\infty}^{z} f(x)\,dx$$

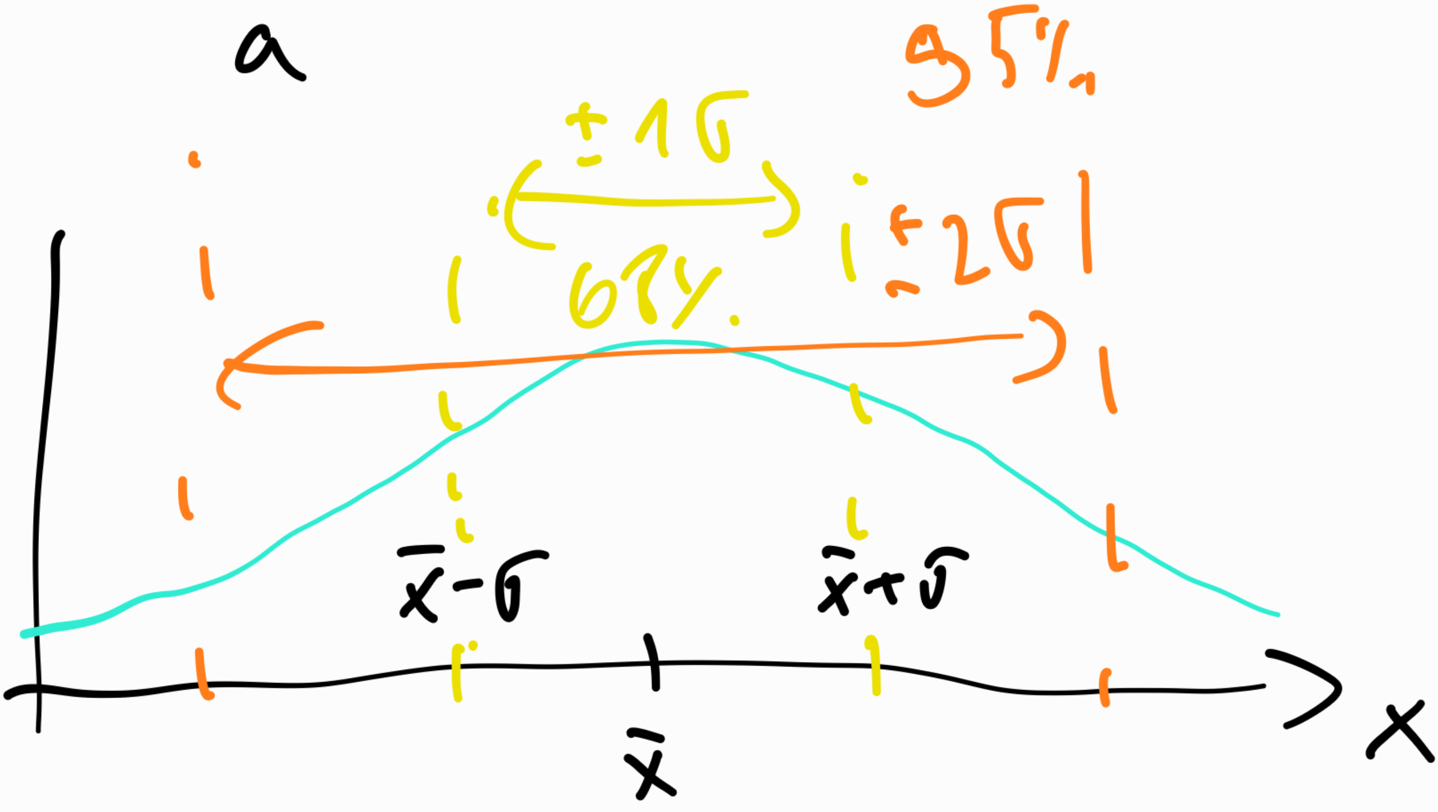
PDF $f(x) \geq 0$

CDF $F(\infty) = 1$

$$F(\infty) = \int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$P(\overset{20}{a} \leq \overset{T}{X} \leq \overset{30}{b}) = \dots =$$

$$= \int_a^b f(x)\,dX$$



$$P(\bar{x}-\sigma \leq x \leq \bar{x}+\sigma) = 68\%$$

$$P(\bar{x}-2\sigma \leq x \leq \bar{x}+2\sigma) = 55\%$$

$$P(\bar{x}-3\sigma \leq x \leq \bar{x}+3\sigma) = 55\%$$

$\bar{x} = \dots$ $\sigma^2 = \dots$